Validation and AI use

Rohan Alwyn | FIN439 Project 1 | Prepared October 19, 2026

Version and evidence boundary

Valuation: September 24 close, using information published by that date. Source pages rechecked October 19. Forecast years end January 31, 2027 through January 26, 2031 under the modeled fiscal-calendar convention. All model figures are USD millions, shares millions, and values/share USD. The first forecast period in DCF is a remaining stub; later periods are annual. Preserve the final repository commit and manifest separately when published.

Source and convention ledger

Input	Date / basis	Source and treatment
FY2026 history and H20	January 25, 2026; filed February 25	NVIDIA 10-K, income/BS/CF tables and gross-margin discussion; reported history retained, normalization separately shown.
Latest opening BS	July 26; released August 26	NVIDIA Q2 FY27 release, condensed BS; assets 320,272, liabilities 91,288, equity 228,984.
H1 / Q3 / shares	H1 actual and Q3 outlook	Q2 release income/EPS/outlook; Q4 is independent forecast judgment; diluted and basic counts kept distinct.
NVIDIA quote	September 24 close, $224.58	FinancialContent historical Close, not Adjusted Close; replaces saved $224.20 intraday reference.
Peer financials	AMD FY2025 / Broadcom FY2025	SEC 10-Ks; annual GAAP EPS/revenue, carrying debt, cash and marketable short-term investments; dated basic share proxies.
Peer prices	September 24 close: AMD 629.26; AVGO 350.36	Stock Analysis historical Close; same date and USD/share.

Exact primary and price URLs are below. The company release is an authoritative issuer source; SEC filings are primary financial statements. Provider price labels are reconciled to Close rather than adjusted total-return prices. No paid API or runtime credential is needed.

Base-year contamination and normalization

Reported FY2026 revenue 215,938, gross profit 153,463 and EBIT 130,387 are retained. Add the 4,500 H20 inventory/purchase-obligation charge only in an analytical normalization column: gross profit 157,963, EBIT 134,887, gross margin 73.15% versus 71.07% reported. This does not restate inventory, cash or equity, and the charge is not added again to forecast FCFF. Other inventory provisions are not all assumed nonrecurring; export-control risk remains in the downside.

Q2/H1 other income includes investment gains; the forecast excludes nonoperating gains and interest income, then values financial assets separately. SBC is treated as cash-equivalent compensation without an addback, which is a conservative economic convention rather than GAAP CFO. The diluted proxy is post-split; pre-split share/EPS fields are not mixed. July debt of 33,366 replaces the old January carrying debt of 8,468. Deferred taxes, leases and other noncurrent accounts are aggregated/fixed, a limitation rather than a solved reconciliation of every subaccount.

Accounting, valuation and failure tests

Claim / risk	Expected outcome	Actual and disposition
Published mini-case	FCFF 77.5m; equity/share 8.20	Matches independent published fixture.
Opening BS	Assets 320,272; liabilities 91,288; equity 228,984	Primary-source totals matched and balanced.
Pro-forma articulation	BS, CF cash, equity, assets, NI, debt and FCFF reconcile	All five future periods pass at 1e-6m tolerance; computed cash is not a balancing plug.

Claim / risk	Expected outcome	Actual and disposition
Timing	Elapsed cash flows excluded from DCF	July-to-September estimates appear only in the rolled opening balances, not as future PV cash flows.
Discount-rate / g boundaries	Higher WACC lowers value; higher g raises value on tested domain	Passed; invalid terminal input rejected. No claim of global monotonicity for every exotic input.
Operating changed input	More inventory investment lowers first-period FCFF	+20 inventory days test passes; this automated result is not the student locked record.
Share bridge	Doubling diluted denominator halves value/share	Passed while basic market-cap weighting held fixed.
Highest-risk missing-flow test	Injecting $1m extra cash must block valuation	Red failure: first forecast BS gap 1.000000m; recovery restores base.
Funding	Unfunded distributions must not create an invented credit line	$1,000 annual dividend stress refused at the cash floor.
Peer definitions	EV = basic market-cap proxy + carrying debt - cash/investments	Both identities pass; annual positive denominators checked; comparability remains qualified.
Restored base	All fields unchanged after automated tests	Exact in the same runtime.
Clean environment	Run from README in fresh venv without network/packages	Passed. First exact-JSON comparison differed across runtimes; maximum numeric difference 4.66e-10, below 1e-6m tolerance. Structural/text comparison matches.

The directional expectations above were encoded before the automated checks in this build. They are AI-authored test specifications, not a student AI-off prediction. Full actual logs: validation-log.txt and cold-run-log.txt. Python standard library only; a fresh bundled-Python venv reproduced the output. The clean run was agent-operated, not falsely attributed to the student.

Assumption challenge register

Driver	Value	Basis	Challenge / evidence that changes it
Q3 / Q4 revenue	108,000 / 118,800m	Q3 guidance; Q4 +10% sequential judgment	Customer spending may slow; update with orders and next results.
Revenue growth FY28-31	35%,25%,15%,8%	Judgment fade as installed base expands	Custom ASICs, export limits, and customer returns may weaken demand; revise with segment results.
Gross margin FY27-31	74%,73%,72%,71%,70%	Near guidance initially; competition/mix fade	Persistent pricing power or cost pressure changes this; verify reported margins and inventory charges.
SG&A / gross profit	2.5%,2.7%,2.9%,3%,3%	Lab operating leverage convention	Costs may track revenue differently; revise on expense mix.
R&D / revenue	8%,9%,10%,11%,12%	Judgment for product and software competition	Compute/cloud and payroll growth may outrun revenue; update on disclosed spending.
Tax rate	17%	Midpoint of 16-18% company FY27 outlook; extended by judgment	Discrete items and geography can change cash tax; separate them from operating tax.
AR / inventory / AP days	64.7 / 129.4 / 61.7	July balances / annualized H1 flow	Not average-balance turnover; seasonality/customer concentration may distort; revise on next filing.

Driver	Value	Basis	Challenge / evidence that changes it
Prepaid / accrued intensity	0.96% / 7.58%	July balance / annualized H1 revenue	All current accruals treated as operating; acquisition/nonrecurring components may require separation.
Depreciation / amortization	23% / 28% of opening assets, time scaled	Simplified asset run-off judgments, informed by prior lab history	Lives/acquisitions can change rates; compare with asset notes and subsequent D&A.
Capex / revenue; split	3%; 97% tangible / 3% intangible	Judgment above prior annual intensity; allocation simplification	Infrastructure expansion may increase investment; revise with cash-flow statements.
Required equity return / debt cost	12% / 4.5%	Judgment ranges; no fitted beta or measured current yield claimed	Equity duration/cyclicality and borrowing yields may change; test 10-14% equity return and recalibrate.
Capital structure weights	Debt 0.61%; remainder equity	Quote x 24.1bn basic shares; reported debt proxy	Debt market value and basic shares can move; recheck outstanding securities.
Terminal growth / reinvestment	3%; stable productive-asset and WC intensity	Nominal steady-state judgment; capital investment funds growth	Margin/duration and capital needs may not converge; use 2-4% grid and compare terminal EV share.
Shares	24,285m diluted; 24,100m basic market-cap proxy	Q2 diluted EPS denominator; prior 10-Q basic cover count	Neither is a forward dilution forecast; revise on awards, issuance and buybacks.
Private investments / haircut	51,157m / 25%	Reported carrying value; liquidity/uncertainty judgment	Impairment, realization tax, concentration and commitments may lower proceeds; independently value portfolio.
Cash floor / dividends	5,000m / $1 annual per basic share	Liquidity judgment; annualized declared quarterly dividend	Liquidity needs and capital returns can change; do not treat policy as guaranteed.
Other balances / acquisitions	Fixed nonoperating balances; no future acquisitions/investments/buybacks	Explicit simplifying exclusions; leases stay in operating expense	Material investments/commitments or capital allocation changes require model revision.
Interim operating roll	Linear July 26 to September 24; first stub prorated	Estimated calendar allocation, not reported interim accounts	Seasonality and capital flows can invalidate the bridge; replace with actuals or a quarterly model.

Partner challenges are not fabricated. The recovered Edition A artifact contains a partner-question label, but this session does not establish a new Weeks 5-6 exchange. Confirm actual attribution if supplying one.

Method disagreement and range dependence

AMD/Broadcom are qualified because of AI compute/networking exposure, not just an industry label. CPUs/client/gaming, software mix, tax items, acquisition amortization, stale basic-share dates and differing fiscal years limit comparisons. Current price times dated shares gives a market-cap proxy; do not call it verified current diluted EV. EV/sales retains operating leases in expenses and excludes preferred/NCI by assumption; it is not EV/EBITDA. P/E uses GAAP annual diluted EPS, not adjusted or forward earnings. Applying peer multiples to NVIDIA's annual historical definitions is mechanical and is not averaged into the DCF.

At +/-5pp revenue growth versus +/-2pp margin, growth produces the larger value span. With growth narrowed to +/-1pp, its span is $10.55 versus $11.45 for margin, reversing the ranking. Sensitivity assigns no probabilities. A growth-only reverse

DCF cannot reach the market inside the admitted +/-10pp domain; fixed base cash flows imply 9.32% WACC to match the $224.58 quote. Do not expand bounds solely to force a convenient answer.

Locked Changed-Input Record

NOT YET SUPPLIED. No completed student-authored locked prediction was located in the recovered Lab 11 evidence. A two-stage student-operated locked_test.py command has been prepared, but the agent has not executed its fresh scenario. Do not substitute the automated tests or a prediction written after seeing output. Preserve any actual record and update this section before freezing the final submission. Missing evidence remains disclosed if not completed.

Named AI-use register

Tool / date	Material contribution	Independent check / disposition
OpenAI Codex, September lab records	Company/peer research, lab code, explanations and sensitivity drafts	Preserved earlier disclosures; personal activities and original checkpoint receipt not inferred.
OpenAI Codex, October 19	Recovered files, read assignment/professor clarification, organized package	Cross-checked Brightspace records and preserved local artifact hash; qualified baseline status.
OpenAI Codex, October 19	Built revised FCFF/stub/WACC/bridge and workbench	Source BS totals, hand-checkable FCFF identity, published fixture and deliberate errors support the correction. Old annual timing and basic-as-diluted interpretation corrected.
OpenAI Codex, October 19	Generated scenario/range/peer analyses, documents and recording preparation	Recomputed from exposed inputs; forecasts and judgment ranges require student review; comparison limitations retained.
OpenAI Codex, October 19	Prepared two-stage locked-test utility	Has not supplied a student prediction or claimed student execution. Actual record is reported only if present above.

The exposed tool identifies a GPT-6-based Codex agent; an exact backend model version is not independently exposed. Material AI assistance is disclosed here rather than repeated as labels throughout the memo. No source or output is accepted merely because a model agrees. The student remains responsible for understanding, reviewing, recording and defending the recommendation. No attendance, partner dialogue, AI-off authorship, on-time checkpoint, video, upload or receipt is invented.

Limits and monitoring

The estimated interim roll, historical balance and share proxies, private-asset values, simplified taxes and constant other balances can materially affect value despite accounting consistency. Three causal cases are not a statistical interval. Rebuild on new guidance, a material customer/export event, capital allocation change, or new filing. The proposed committee action is watch/defer; do not initiate on model arithmetic alone. Compare new price/value to the disclosed 20% margin-of-safety gate and reconsider risk at each review.

Source URLs

- NVIDIA 10-K: https://www.sec.gov/Archives/edgar/data/1045810/000104581026000021/nvda-20260125.htm

- NVIDIA Q2 release: https://investor.nvidia.com/news/press-release-details/2026/NVIDIA-Announces-Financial-Results-for-Second-Quarter-Fiscal-2027/

- NVIDIA historical price: https://www.financialcontent.com/quote/NQ:NVDA/historical

- AMD 10-K: https://www.sec.gov/Archives/edgar/data/2488/000000248826000018/amd-20251227.htm

- Broadcom 10-K: https://www.sec.gov/Archives/edgar/data/1730168/000173016825000121/avgo-20251102.htm

- Peer prices: https://stockanalysis.com/stocks/amd/history/ and https://stockanalysis.com/stocks/avgo/history/

- Project: https://github.com/CinderZhang/FIN43900-Fall2026/blob/main/projects/project-1-corporate-finance-application.md